Krittin Chaowakarn

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EDUCATION

Bachelor of Engineering in Electrical Engineering — 08/2021–11/2025
Sirindhorn International Institute of Technology, Thammasat University — Pathum Thani, Thailand

- Relevant coursework: Linear Algebra, Robotics, Control Systems, Probability and Random Processes
- GPA: 3.87/4.00 – **First-Class Honors**

TUMexchange (Exchange Semester) — 04/2025–09/2025
Technical University of Munich — Munich, Germany

- Thesis: *"Real-Time Object Detection for Autonomous Driving: An Empirical Study in a Small-Scale Urban Environment"* under the supervision of Prof. Andreas Herkersdorf
- Relevant coursework: Multimodal Deep Learning, Software Engineering, Bachelor's Thesis

PREPRINTS

K. Chaowakarn, P. Sangwongngam, N. H. H. Aung, and C. Charoenlarpnopparut, "*NV3D: Leveraging Spatial Shape Through Normal Vector-based 3D Object Detection*," *arXiv preprint* arXiv:2510.11632, 2025. [Online]. Available: `https://arxiv.org/abs/2510.11632`

RESEARCH EXPERIENCE

Bachelor's Thesis Student — 04/2025–09/2025
Technical University of Munich — Munich, Germany
Advisor: Prof. Andreas Herkersdorf

- Collected and curated an image dataset from the lab environment using k-means clustering on L2-normalized embeddings (unit hypersphere representation), reducing ∼1,000 raw images to 119 distinct samples
- Conducted an empirical study on YOLO-based real-time object detection in a small-scale urban environment using low-power NVIDIA Jetson Nano, evaluating latency, throughput, and hardware utilization

Undergraduate Researcher / Research Intern — 06/2023–04/2025
National Electronics and Computer Technology Center — Pathum Thani, Thailand
Advisors: Dr. Paramin Sangwongnam and Assoc. Prof. Chalie Charoenlarpnopparut

- Conducted research on LiDAR-based 3D multi-object detection for autonomous vehicles and proposed two normal-vector-guided sampling strategies that reduced redundant points by up to 55%
- Accelerated normal vector computation by 6.67× through CUDA C++ parallelization, reducing runtime from 0.10 s (PyTorch) to 0.015 s

Technical Collaborator — 01/2025–03/2025
Sirindhorn International Institute of Technology, Thammasat University — Pathum Thani, Thailand
Advisor: Assoc. Prof. Itthisek Nilkhamhang

- Contributed to multi-robot formation research using Yahboom robots and ROS 2 simulations
- Deployed a vision-based target tracking system integrating YOLO (detection), DeepSORT (tracking), and face recognition (identity verification) for human-following robots

Research Intern — 06/2022–07/2022
National Electronics and Computer Technology Center — Pathum Thani, Thailand
Advisor: Dr. Paramin Sangwongnam

- Performed a deep feedforward neural network using TensorFlow for mmWave beam and blockage prediction utilizing Sub-6 GHz signals

TEACHING/ACADEMIC EXPERIENCE

Teaching Assistant 01/2023–Present
Sirindhorn International Institute of Technology, Thammasat University Pathum Thani, Thailand
*Teaching Assistant for 9 courses: 5 grading roles, 1 lab TA, and 3 upcoming lab TA**

- Control Systems Laboratory (01/2026–05/2026)*
- Electronics and Microelectronics Laboratories (01/2026–05/2026)*
- Electrical Engineering Crafting Skill (01/2026–05/2026)*
- Electromagnetics (01/2024–05/2024, 01/2025–05/2025)
- Linear Algebra and Optimization Method (08/2024–12/2024)
- Digital Circuits Laboratory (08/2024–12/2024)
- Computational Tools in Electrical Engineering (08/2023–12/2023)
- Basic Electrical Engineering (01/2023–05/2023)

Invited Instructor 01/2026
National Electronics and Computer Technology Center Pathum Thani, Thailand

- Delivered a three-day training program on machine learning and deep learning for undergraduate interns

WORK EXPERIENCE

AI Engineer Intern 06/2024–07/2024
BOTNOI Group Bangkok, Thailand

- Improved Thai SNOMED-CT medical vocabulary using a machine translation model, achieving over 80% reduction in translation ambiguities (70 out of 86 issues solved from a dataset of 385)
- Developed a greedy, embedding-based alignment algorithm for English-Thai (unsegmented) text to build genre-specific MT datasets, cutting manual matching time by 80% (50 to 10 hours)

HONORS AND AWARDS

Young Scientist and Technologist Program 08/2021–05/2025

- Full scholarship from the National Science and Technology Development Agency; 1 of 4 awardees (∼0.25%)

Top 100 Team – KPIT Sparkle 2024 04/2024

- Selected among the top 100 teams (∼0.45%) worldwide in KPIT Sparkle 2024, a global student innovation contest focused on vehicle technologies
- Proposed a risk assessment system for surrounding vehicles based on their driving behaviors, helping a driver with situational awareness and assisting insurance companies in evaluation

Keio University International Workshop 2023 11/2023

- Selected as one of 12 institute representatives to attend a workshop at Keio University, participating in laboratory activities and cultural exchange with Japanese students

EXTRACURRICULAR ACTIVITIES

Vice President of the Electrical Engineering Students Council 01/2023–05/2025
Sirindhorn International Institute of Technology, Thammasat University Pathum Thani, Thailand

AI/ML Team Lead 10/2023–05/2024
Google Developer Student Club, Thammasat University Pathum Thani, Thailand

- Presented a keynote workshop on *Introduction to Machine Learning and Its Applications* to an audience of 35 undergraduate participants

SKILLS

Programming: Python, C/C++, Linux (CLI, Bash), Git

Languages: English (professional proficiency, TOEFL iBT 100); Thai (native); German (basic)